

CATALOG
IIT GUWAHATI
DEPARTMENT OF CSE
Mid Semester Examination

CS207 ALGORITHMS

FEBRUARY 25, 2024

Time: 09:00am-11:00am

Max Marks: 70

Name

Sign

Roll

Invigilator

INSTRUCTIONS

- This question paper has **10** pages and **27** questions.
- The answers to the 25 MCQs(2 marks each) are to be bubbled on the Answer Sheet.
- The answers to the 2 long answer type questions (10 marks each) are to be written only in the space provided for the same. These questions will be used only to determine if you are eligible for AS, AA and AB grades.
- Your UID is the last 3 digits of your roll number.
- You may use the supplementary sheets for the rough work. Do not attach your rough work/ supplementary sheets.
- It is suggested that you use a ball point pen. Black is preferred.

MULTIPLE CHOICE QUESTIONS

Question 1 [MST01] Which of the following statements are true regarding Kruskal's algorithm on a graph G ? Assume that the edge weights are distinct.

- ☐ A The algorithm always discards the heaviest edge of G .
☒ B The algorithm will always pick the lightest edge of G .
☐ C The algorithm will never pick all the edges of G .
☐ D The algorithm runs in $O(n)$.

Question 2 [MST02] Which of the following statements are true regarding Minimal Spanning Trees?

- ☐ A The minimal spanning tree is unique in all graphs.
☒ B In a graph with distinct positive edge weights, there is a unique minimal spanning tree.
☒ C The heaviest edge in a cycle is absent in some minimal spanning tree.
☒ D The lightest edge of some cut is present in some minimal spanning tree.

Question 3 [MST03] Consider a weighted complete graph G on vertex set $\{v_1, v_2, \dots, v_n\}$ such that the weight of the edge (v_i, v_j) is $(i - j)^2$. The weight of the minimum spanning tree of G is:

- ☐ A $(n - 1)^2$ ☒ B $n - 1$ ☐ C $\binom{n}{2}$ ☐ D n^2

Question 4 [MST04] What is the best algorithm to check whether a cycle exists while running Kruskal's algorithm?

- ☐ A DFS ☒ B Union Find Algorithm ☐ C BFS ☐ D Kosaraju's Algorithm

Question 5 [MST07] Let G be a weighted graph on 4 vertices. What is the maximum number of minimal spanning trees that G can have?

- ☐ A 24 ☒ B 16 ☐ C 6 ☐ D 1 ☐ E 4

Question 6 [CUT01] What is the number of cuts in a graph with n vertices?

- ☐ $n - 1$ ☐ 2^{n-1} ☐ $\binom{n}{2}$ ☒ $2^{n-1} - 1$

Question 7 [MST05] Which data structure is more appropriate for finding the nearest neighbor in Prim's algorithm?

- ☒ Priority Queue ☐ Adjacency Matrix ☐ BST ☐ Adjacency List

Question 8 [MST06] Let G be a weighted graph with positive edge weights. Let w be the weight of its minimal spanning tree. Let G_1 be the graph obtained by doubling the edge weights of 99.999% of the edges. Let w_1 be the weight of a minimal spanning tree of G_1 . Which of the following statements are true?

- ☒ w is unique. ☐ $w_1 \neq w$.
☐ w is unique only if edge weights are unique. ☒ If G was K_{200} then $w_1 \neq w$.

Question 9 [FFT01] Let $A(x)$ be a linear polynomial over $\mathbb{R}$. Let $B(x)$ be a quadratic polynomial over $\mathbb{R}$. Let $c_0 + c_1x + c_2x^2 + c_3x^3$ be the product of $A(x)$ and $B(x)$. Further, $A(1)B(1) = 24$, $A(-1)B(-1) = -2$, $A(i)B(i) = -9 - 3i$ and $A(-i)B(-i) = -9 + 3i$. What is (c_0, c_2) ?

- ☒ $(1, 10)$ ☐ $(1, 5)$ ☐ $(8, 5)$ ☐ $(10, 1)$

Question 10 [FFT02] Let $A(x)$ be a linear polynomial over $\mathbb{R}$. Let $B(x)$ be a quadratic polynomial over $\mathbb{R}$. Let $c_0 + c_1x + c_2x^2 + c_3x^3$ be the product of $A(x)$ and $B(x)$. Further, $A(1)B(1) = 24$, $A(-1)B(-1) = -2$, $A(i)B(i) = -9 - 3i$ and $A(-i)B(-i) = -9 + 3i$. What is (c_1, c_3) ?

- ☒ $(5, 8)$ ☐ $(10, 8)$ ☐ $(8, 5)$ ☐ $(1, 10)$

Question 11 [HUFF01] What is the primary purpose of using Huffman encoding scheme?

- ☐ It guarantees data encryption for security purposes.
☐ It significantly increases the speed of data transmission.
☐ It uses fixed length codes there by simplifying the decoding.
☒ It reduces the overall size of the data.

Question 12 [HUFF02] Consider the following statements regarding Huffman encoding;

S1: A more frequent symbol is always assigned a strictly shorter binary code.

S2: A more frequent symbol is never assigned a strictly longer binary code.

Which of these statements are true?

- ☒ Only S2. ☐ Both ☐ None ☐ Only S1.

Question 13 [SB01] Consider the following preference lists for three women (w_1, w_2, w_3) and three men (m_1, m_2, m_3) :

$$\begin{array}{ll} m_1 : w_1 > w_2 > w_3 & w_1 : m_2 > m_1 > m_3 \\ m_2 : w_2 > w_1 > w_3 & w_2 : m_1 > m_2 > m_3 \\ m_3 : w_1 > w_2 > w_3 & w_3 : m_1 > m_2 > m_3 \end{array}$$

Which pair is unstable in the matching $\{m_1 - w_1, m_2 - w_3, m_3 - w_2\}$

- ☐ $m_1 - w_2$ ☒ $m_2 - w_1$ ☐ $m_2 - w_3$ ☐ $m_3 - w_1$ ☐ $m_1 - w_1$

Question 14 [SB02] Suppose that in a matching M , (m, w) is an unstable pair. Is it possible that the Gale Shapley Algorithm produces a matching in which m and w are paired?

- ☐ Yes ☒ No ☐ Insufficient data.

Question 15 [SB03] What is a stable matching in the Stable Marriage Problem?

- ☐ A matching where everyone is perfectly happy.
☐ A matching with the maximum number of marriages.
☒ A matching with no unstable pairs.
☐ A random pairing of individuals.

Question 16 [SB04] What is the time complexity of the Gale Shapley algorithm with n men and n women?

- ☐ $O(n \log n)$ ☒ $O(n^2)$ ☐ $O(n^3)$ ☐ $O(2^n)$

Question 17 [SB05] Suppose that the data corresponding to a stable marriage problem instance is 5MB in size. Let the time taken by the Gale Shapley algorithm on the worst such input instance be 5 seconds. How much time (roughly) will the Gale Shapley Algorithm take to process the worst input of size 50MB?

- ☐ 5s ☒ 50s ☐ 500s ☐ 5000s

Question 18 [MED01] What is the recurrence relation for the divide and conquer algorithm to find the k^{th} smallest element in an array of n elements using the median of medians algorithm discussed in class?

- ☐ $T(n) = T(n/5) + 7T(n/10) + cn$ ☐ $T(n) = T(7n/10) + T(3n/10) + cn$
☐ $T(n) = T(n/5) + T(4n/5) + cn$ ☒ $T(n) = T(n/5) + T(7n/10) + cn$

Question 19 [MED02] Which of the following statement(s) is/are true regarding the algorithm we saw in class for finding k^{th} smallest element of an array A with n elements?

- ☒ The pivot selection guarantees that there are at least $\frac{n}{5}$ elements in A which is less than pivot.
☒ The pivot selection guarantees that there are at least $\frac{3n}{10}$ elements in A which is greater than pivot.
☒ The pivot selection guarantees that there are at least $\frac{3n}{10}$ elements in A which is less than pivot.
☒ The pivot selection guarantees that the cardinality of elements that are lesser than pivot or the cardinality of elements that are equal to pivot or the cardinality of elements that are greater than pivot is at most $\frac{7n}{10}$

Question 20 [COIN01] We are given n coins. There are 2 bad coins(having larger weights than others). If the task is to identify those two bad coins, what can you say about the cardinality of the search space?

- ☐ $4n$ ☐ $O(n)$ ☒ $n(n-1)/2$ ☐ $n(n-1)$

Question 21 [COIN02] We are given n coins. There are 2 bad coins(having different weights than others). If the task is to identify those two bad coins and the type of fault, what can you say about the cardinality of the search space?

- ☐ $4n$ ☐ $4n(n-1)$ ☒ $2n(n-1)$ ☐ $n(n-1)$

Question 22 [COIN03]

We are given n coins. There are 3 bad coins(having different weights than others). The task is to identify those three bad coins and the type of fault. You are given a beam balance. Let k be the number of times the beam balance has to be used in the worst case for an optimal algorithm. Which of the following statements are true?

- ☐ $\binom{n}{3} \geq 3^k$ ☐ $\binom{n}{3} \geq 2^k$ ☒ $\binom{n}{3} \times 8 \leq 3^k$ ☒ k is $O(\log n)$

Question 23 [CP01] Which of the following statement(s) is/are true regarding the algorithm we saw in class for finding closest pair of points given n points in $2D$ region? Let δ_L denotes the minimum distance between closest pair of points in left half, δ_R denotes the minimum distance between closest pair of points in right half and δ denotes minimum of δ_L and δ_R .

- ☐ A Every point in the 2δ band whose distance to a point p is less than δ , lies in the $2\delta \times \delta$ rectangle of the point p .
- ☒ B Every point in the 2δ band whose distance to a point p is less than δ and is below to point p lies in the $2\delta \times \delta$ rectangle of the point p .
- ☒ C The number of candidate points (distance to p less than δ) to look for point p in the 2δ band is at most 8.
- ☐ D The number of candidate points (distance to p less than δ) to look for point p in the 2δ band is at most 3.

Question 24 [MASTER02] Solve the following recurrence for $T(n)$. $T(n) = 64T(n/4) + n$

- ☐ A $\Theta(n^3 \log n)$ ☐ B $\Theta(n^{16})$ ☒ C $\Theta(n^3)$ ☐ D $\Theta(n^{64})$

Question 25 [MASTER01] Solve the following recurrence for $T(n)$. $T(n) = 2T(n/2) + n \log n$

- ☐ A $\Theta(n \log n)$ ☐ B $\Theta(n \log(n^2))$ ☒ C $\Theta(n \log^2 n)$ ☐ D $\Theta(n^2)$

LONG ANSWER QUESTION

D1. Suppose you are given two sets of n points, one set $\{p_1, p_2, \dots, p_n\}$ on the line $y = 0$ and the other set $\{q_1, q_2, \dots, q_n\}$ on the line $y = 1$. Consider the set of n line segments formed by connecting each point p_i to the corresponding point q_i . Describe and analyze an algorithm to determine how many pairs of these line segments intersect.

D2. Let G be a weighted graph with distinct positive edge weights. Let G' be the graph obtained by replacing the edge weights by their logarithm. Does Kruskal's algorithm produce the MST correctly for G' ? Justify your answers.

— SPACE FOR LONG ANSWER QUESTIONS —

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ANSWER SHEET

Get the invigilator's help to know your UID. Invigilator's Sign :

STUDENT INFORMATION

Please bubble your UID.

Name

0 1 2 3 4 5 6 7 8 9

UID.....Roll

0 1 2 3 4 5 6 7 8 9

Email.....

0 1 2 3 4 5 6 7 8 9

RESPONSES

Q 1: A C D

Q 14: A C

Q 2: A

Q 15: A B D

Q 3: A C D

Q 16: A C D

Q 4: A C D

Q 17: A C D

Q 5: A C D E

Q 18: A B C

Q 6: A B C

Q 19:

Q 7: B C D

Q 20: A B D

Q 8: B C

Q 21: A B D

Q 9: B C D

Q 22: A B

Q 10: B C D

Q 23: A D

Q 11: A B C

Q 24: A B D

Q 12: B C D

Q 25: A B D

Q 13: A C D E

PLEASE DO NOT WRITE ANYTHING ON THE OTHER
SIDE OF THE ANSWER SHEET.

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